

Victor Santana

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RELEVANT SKILLS

- **Programming (in order of proficiency):** C#, C++, Python, Java, C, Rust, HTML/CSS, Node.js, Assembly
- **Embedded & Hardware:** Arduino, STM32, KiCad, Altium, LTSpice, TinkerCad, Logisim Evolution, FPGA, HLS, Verilog/SystemVerilog
- **Lab Equipment:** Oscilloscopes, Function Generators, Digital Multimeters, Soldering
- **Tools & Platforms:** Git/GitHub, VS Code, API Integration, AI-Assisted Development, OOP, Serial Communication, Networking, Linux (Ubuntu), Vitis/Vivado, Docker, Make/CMake
- **Leadership & Languages:** Professional Written & Oral Communication, Team Management, Technical Training, Mentoring, Spanish

EDUCATION

Bachelor of Science in Computer Engineering

Anticipated: July 2027

Brigham Young University - Idaho

Rexburg, Idaho

- **Certificates:** Computer Programming & Embedded Systems
- **GPA:** 4.0/4.0
- **Scholarship:** Brigham Young University-Idaho Grant 2024-2027 (Acknowledgement of Academic Excellence)
- **IEEE Student Branch Vice Chair:** Directed multi-disciplinary engineering teams, orchestrated hands-on technical workshops, and executed industry networking events

EXPERIENCE

Undergraduate Research Assistant (IMMERSE-X)

May 2026 - Present

Brigham Young University

Provo, Utah

- Engineered an automated, 6-script toolchain using Docker and QEMU to generate reproducible PYNQ/PetaLinux OS images for the AMD AUP-ZU3 board
- Developed custom Python I2C telemetry scripts to read raw, memory-mapped silicon registers directly from the OV5640 image sensor and Xilinx FPGA D-PHY IPs
- Utilized oscilloscopes to physically verify high-speed MIPI CSI-2 camera interfaces on the AUP-ZU3 board. Probed differential clock lanes of Pcam 5C modules to definitively validate the hardware failure hypothesis
- Provided a comprehensive technical presentation at IMMERSE 2026 detailing the Zynq UltraScale+ architecture, PYNQ image automation, and 2 MIPI camera debugging methodologies to a technical audience

Faculty Technology Center Lead

May 2024 - May 2026

Brigham Young University - Idaho

Rexburg, Idaho

- Led and coordinated a 20+ member team, managing task delegation and one-on-one technical training to achieve great appointment resolution rate
- Exceeded weekly deployment targets by 120% by automating setup workflows, delivering 10+ complete 14-week courses weekly via Canvas
- Improved and maintained 100+ online courses, applying systematic problem-solving to resolve configuration issues and optimize end-user experience

PROJECTS

- **Smart RC Car | BYU STEM Camp Team Project (May 2026 - Present):** Engineered an ESP32-CAM Wi-Fi web server and motor control framework, featuring a WebSocket-driven mobile dashboard to provide sub-millisecond MJPEG video streaming and real-time hardware tuning for an educational workshop
- **Tinaco Smart Water Level Monitor | Personal (Feb 2026):** Designed an Arduino and C++ ultrasonic monitoring system to computerize water tracking, delivering a real-time capacity readout via an I2C LCD and LED alerts, eliminating manual rooftop inspections